1.textbook,Radio buttons,Checkbooks

```
<!DOCTYPE html>
<html>
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>Form Validation</title>
   <script>
      function validationForm() {
          var email = document.forms["myForm"]["email"].value;
          var mobile = document.forms["myForm"]["mobile"].value;
          var name = document.forms["myForm"]["name"].value;
         var emailRegex = /^[a-zA-A0-9._-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,4}$/;
         var mobileRegex = /^[0-9]{10}$/;

         if(name === "") {
            alert("Name must be filled out");
            return false;
         }

         if(!emailRegex.test(email)) {
            alert("Invalid email format");
            return false;
         }

         if(!mobileRegex.test(mobile)) {
            alert("Mobile number must be 10 digits");
            return false;
         }
      }
   </script>
</head>
<body>
```

```
    <form name="myForm" onsubmit="validationForm();" method="post">
        Name: <input type="text" name="name" autocomplete="on"><br><br>
        Email: <input type="text" name="email" autocomplete="on"><br><br>
        Mobile: <input type="text" name="mobile"><br><br>
        Gender:
        <input type="radio" name="Gender" value="male">Male
        <input type="radio" name="Gender" value="female">Female <br><br>
        Hobbies:
        <input type="checkbox" name="hobby1" value="Reading">Reading
        <input type="checkbox" name="hobby2" value="Travelling">Travelling <br><br>
        <input type="submit" value="Submit">
    </form>
</body>
</html>
```

2.Develop an html form which accepts any mathematical expression.write javascript code to evaluate the expression and display the result.

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Math Expression Evaluator</title>
</head>
<body>
    <form id="mathForm">
        <label for="expression">Enter Mathematical Expression:</label>
        <input type="text" name="expression"    id="expression">
        <input type="button" value="Evaluate" onclick="evaluateExpression()">
     </form>
    <div id="result"></div>
    <script>
        function evaluateExpression()
        {
```

```
 try{
            var expression = document.getElementById('expression').value;
            var result = eval(expression);
            document.getElementById('result').innerText='Result:'+result;
        }
        catch(error)
        {
            document.getElementById('result').innerText="Invalid Expression";
        }
      }
  </script>
</body>
</html>
```

3.create a page with dynamic effects.Write the code to include layers and basic animation.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Dynamic Page with Animation</title>
  <style>
    .layer{
      width: 200px;
      height: 200px;
      background-color: skyblue;
      position: absolute;
      text-align: center;
      line-height: 200px;
      border-radius: 10px;
      left: 0;
      top: 100px;
    }
  </style>
```

```
</head>
<body>
  <div id="movingLayer" class="layer">Web Programming</div>
  <script>
    document.addEventListener('DOMContentLoaded',() =>{
      const layer=document.getElementById('movingLayer');
      let position=0;
      let direction =1;

      setInterval(()=>{
        if (position>window.innerWidth-200 || position<0)
        {
          direction*=-1;
        }
        position +=5*direction;
        layer.style.left=position+'px';
      },50);
    });
  </script>
</body>
</html>
```

4.Write a Java Script code to find the sum of N natural Numbers.(User user-defined function)

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Sum of Natural Numbers</title>
  <script>
    function sumOfNaturalNumbers(n){
      let sum=0;
      for (let i=1;i<=n;i++){
        sum+=i;
```

```
        }
        return sum;



    }

    function calculateSum()
    {
        let n=document.getElementById("numberInput").value;
        n=parseInt(n);
        if(!isNaN(n)&&n>0)
        {
            let sum =sumOfNaturalNumbers(n);
            alert("Sum of first "+n+" natural number is:"+sum);
        }       else{
            alert("Please enter a positive integer");
        }
    }
  </script>
</head>
<body>
  <h2>Sum of N Natural Numbers</h2>
  <label for="numberInput">Enter a number</label>
  <input type="number" id="numberInput" min="1">
  <button onclick="calculateSum()">calculateSum</button>
</body>
</html>
```

5.write a JavaScript code block using arrays and generate the current date in words.

```
<!DOCTYPE html> <html lang="en">
<head>
  <meta charset="UTF-8">
```

```
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>Current date in words</title>
</head>
<body>
   <h1>Today's Date</h1>
   <p id="dateInWords"></p>
   <script>        const
days=["Sunday","Monday","Tuesday","Wednesday","Thursday","Friday","Saturday"];
   const
months=["January","February","March","April","May","June","July","August","September",
"October","November","December"];
      const now = new Date();

      const dayName= days[now.getDay()];
      const monthName = months[now.getMonth()];
      const year = now.getFullYear();
      const date = now.getDate();

      const dateString = `${dayName},${monthName} ${date},${year}`;

      document.getElementById("dateInWords").innerText=dateString;
      </script>
</body>
</html>
```

6.**Create a** form **student information.Write a javascript code top find Total,Average,Result and Grade.**

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>Student Information</title>
   <script>
```

```
        function calculateResults(){
          var marks = document.getElementsByClassName('marks');
          var total =0;
          for(var i=0;i<marks.length; i++)
          {
            total += parseInt(marks[i].value,10);
          }
          var average = total/marks.length;
          var result = average >=40? 'Pass':'Fail';
          var grade ='';

          if(average>=80)
          {
            grade='A';
          }
          else if(average>=60)
          {
            grade='B';
          }
          else if(average>=40)
          {
            grade='C';
          }        else{
            grade='F';
          }

          document.getElementById('total').innerHTML='Total: '+total;
          document.getElementById('average').innerHTML='Average: '+average.toFixed(2);
          document.getElementById('result').innerHTML='Result: '+result;
          document.getElementById('grade').innerHTML='Grade: '+grade;
        }
      </script>
    </head>
```

```
<body>
  <h2>Student Information</h2>
  <form>
    Name: <input type="text"  id="name"> <br><br>
    Marks 1: <input type="number" class="marks"> <br><br>
    Marks 2: <input type="number" class="marks"> <br><br>
    Marks 3: <input type="number" class="marks"> <br><br>

    <button type="button" onclick="calculateResults()">Calculate</button>
  </form>

  <p id="total"></p>
  <p id="average"></p>
  <p id="result"></p>
  <p id="grade"></p>
</body>
</html>
```

8.background color based on day of week

```
<!DOCTYPE html>
<html>
  <head>
    <title>Color Change by Day</title>
    <?php
    $daysOfWeek = array(
      'Sunday'=>'#FF5733',
      'Monday'=>'#33FF57',
      'Tuesday'=>'#3357FF',
      'Wednesday'=>'#FF33FF',
      'Thursday'=>'#FF33FF',
      'Friday'=>'#33FFFF',
      'Saturday'=>'#FF3333'
```

```
    );

    $currentDay=date('l');
    $backgroundColor='#FFFFFF';

    if(array_key_exists($currentDay,$daysOfWeek))
    {
      $backgroundColor=$daysOfWeek[$currentDay];
    }

    ?>

    <style>          body{
          background-color:<?php echo $backgroundColor;?>;
      }
    </style>
  </head>
  <body>
    <h1>Welcome! Today is <?php echo $currentDay;?>.</h1>
    </body>
  </html>
```

9.Prime Numbers and Fibonacci series.

```
<?php
function generatePrimes($maxNumber)
{
  echo"Prime numbers upto $maxNumber: ";
  for($number=2;$number<=$maxNumber;$number++)
  {
    $isPrime=true;
    for($i=2;$i<=sqrt($number);$i++)
    {
      if($number%$i==0)
      {
```

```
                $isPrime=false;
                break;
            }
        }
        if($isPrime)
        {
            echo $number. " ";
        }
    }
        echo "<br>";
}

function fibonacciSeries($numTerms)
{
    $first =0;
    $second=1;

    echo "Fibonacci Series($numTerms terms):";
    for($i=0;$i<$numTerms;$i++)
    {
        echo $first . " ";
        $next =$first+$second;
        $first=$second;
        $second=$next;
    }
    echo "<br>";
}

generatePrimes(50); fibonacciSeries(10);
```

10.Remove Duplicates from Sorted list.

```
<?php
function removeDuplicates($array)
{
```

```
    $result=array_values(array_unique($array));
    return $result;
}

$sortedList =[1,1,2,2,3,3,4,5,5];
$uniqueList = removeDuplicates($sortedList);

echo"Original List: ";
print_r($sortedList);
echo"<br> Unique List: ";
print_r($uniqueList);

?>
```

11.write a php Script to following pattern on the screen.

```
<?php $rows =5;
for($i=$rows;$i>0;$i--)
{
    for($j=$rows-$i;$j>0;$j--)
    {
        echo" ";
    }
    for($k=0;$k<$i;$k++)
    {
        echo "*";
    }
    echo "<br>";
}
?>
```

12.Write a simple program in PHP for searching of data by different criteria.

```
<?php

$data=[
```

```
    ['id'=>1,'name'=>'Srikanth','age'=>30],
    ['id'=>2,'name'=>'Srikanth','age'=>35],
    ['id'=>3,'name'=>'Srinivas','age'=>50],
    ['id'=>4,'name'=>'Smayan','age'=>45],
    ['id'=>5,'name'=>'Saatvik','age'=>50]
];

function searchByCriteria($data, $criteria)
{
    $results = [];

    foreach ($data as $entry)
    {
        $match= true;
        foreach($criteria as $key => $value)
        {
            if(!isset($entry[$key])|| $entry[$key] !=$value)
            {
                $match=false;
                break;
            }
        }
        if($match)
        {
            $results[]=$entry;
        }    }
    return $results;
}

$criteria =['age' => 50];

$searchResults = searchByCriteria($data,$criteria);
```

```
echo "Search Results:<br>";
print_r($searchResults);

?>
```

13.Generate Captcha

```
<?php
session_start();

function generateCaptchaCode($length=6)
{
   $characters='0123456789abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUV WXYZ';
   $charactersLength=strlen($characters);
   $randomString=' ';    for($i=0;$i<$length;$i++)
   {
      $randomString .=$characters[rand(0,$charactersLength -1)];
  }
   return $randomString;
}

echo "Captcha Code is: ".generateCaptchaCode();
?>
```

17.validate input.

```
<!DOCTYPE html>
<html lang="en">
<head>
   <title>Input Validation</title>
</head>
<body>
   <form method="post">
      Name: <input type="text" name="name"><br><br>
      Email: <input type="text" name="email"><br><br>
      <input type="submit" name="submit" value="Submit">
   </form>
</body>
```

```
</html>

<?php
if($_SERVER["REQUEST_METHOD"] =="POST")
{
   $name = $_POST['name'];
   $email = $_POST['email'];

   if(empty($name))
   {
      echo "Name is required.<br><br>";
   }    else{
      echo "Name: ".$name."<br><br>";
   }
   if(empty($email))
   {
      echo "Email is required.<br><br>";

   }elseif(!filter_var($email,FILTER_VALIDATE_EMAIL))
   {
      echo "Invalid email format.<br><br>";
   }    else{
      echo "Email: " .$email ."<br><br>";
   }
}
?>
```

18.Setting and Retrieving a cookie.

```
<!DOCTYPE html>
<html lang="en">
<head>
   <title>Cookie Handler</title>
</head>
<body>
   <?php
```

```
    if($_SERVER["REQUEST_METHOD"]=="POST")
    {
        setcookie("username",$_POST['username'],time()+(10*30),"/");
        echo "<p>Cookie set. Reload the page to see the cookie value.</p>";
    }

    if(isset($_COOKIE["username"]))
    {
        echo "<p>Welcome back, " . htmlspecialchars($_COOKIE["username"]). "!</p>";
    }    else{
        echo "<p>Welcome, guest! </p>";
    }
    ?>

    <form method="post" action="<?php echo htmlspecialchars($_SERVER["PHP_SELF"]);?>">
        <label for="username">Enter your name:</label><br>
        <input type="text" id="username" name="username"><br>
        <input type="submit" value="Set Cookie">
    </form>
</body>
</html>
```

20.Exception handling.

```
<?php
function divide($numerator, $denominator)
{
    if($denominator == 0)
    {

        throw new Exception("Cannot divide by zero.");
    }
    return $numerator/ $denominator;
}

function checkDateFormat($date, $format = 'Y-m-d') {
```

```
    $dateTime = DateTime::createFromFormat($format,$date);
    if(!$dateTime || $dateTime->format($format) != $date)
    {
        throw new Exception("Invalid date format.");
    }
    echo "The date" . $date . "is valid. <br>";
    return true;
}

try{
    echo divide(10,2) . "<br>";
    echo divide(10,0) . "<br>";
}
catch (Exception $e)
{
    echo "Division error: " . $e->getMessage() . "<br>"; }

try{
    checkDateFormat("2023-03-10");
    checkDateFormat("10/03/2023");
}
catch(Exception $e)
{
    echo "Date error: " .$e->getMessage() . "<br>";
}

?>
```